

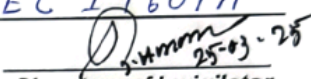
Course B-Tech Semester Examination, 20 24 - 20 25

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Signature of Invigilator

Group - A

1. (i)

$$\Delta e^x = e^{x+h} - e^x$$
$$= e^x \cdot e^h - e^x$$

$$\Delta e^x = (e^h - 1) e^x$$

$$\therefore \Delta^3 e^x = (e^h - 1)^3 e^x$$

putting $h = 1$

$$\Delta^3 e^x = (e - 1)^3 e^x$$

(ii) The degree of precision in Trapezoidal Rule is 1

(iii) Extrapolation is the process of estimating values of a function beyond the known datapoints using an approximation algorithm.

(iv) An example of ~~Truncat~~ Truncation error can be the Maclaurian series of e .

$$e = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

where the error is $\rightarrow \frac{x}{n!} + \frac{x}{n!}$

(v) We use Newtons Forward Interpolation Formula when the datapoints are spaced equally and interpolation is required near the beginning.

Group-B

2. (i) we know,

$$\Delta f(x) = f(x+h) - f(x) \quad \text{--- (i)}$$

taking log on $f(x)$ i.e. $\log f(x)$

$$\begin{aligned} \Delta \log f(x) &= \log(f(x+h)) - \log f(x) \\ &= \log \left[\frac{f(x+h)}{f(x)} \right] \end{aligned}$$

$$\text{From (i)} \quad f(x+h) = \Delta f(x) + f(x)$$

$$\therefore \Delta \log f(x) = \log \left[\frac{\Delta f(x) + f(x)}{f(x)} \right]$$

$$\Delta \log f(x) = \log \left[1 + \frac{\Delta f(x)}{f(x)} \right]$$

$$(ii) E^{1/2} = \mu + \frac{1}{2} \delta$$

shift operator is given by.

$$\text{E} f(x) = f(x+h)$$

~~we know the central difference operator is~~

$$\mu = \frac{f(x+h) + f(x-h)}{2}$$

Square root operator

$$E^{1/2} = 1 + \frac{1}{2} \delta$$

where δ is the central different operator.

$$\Rightarrow E^{1/2} = \mu + \frac{1}{2} \delta$$

4)

x	y
3	11
7	17
9	24
12	30

we know,

$$f(x) = y_1 \frac{(x-x_1)(x-x_2)}{(x_1-x_2)(x_1-x_3)} + y_2 \frac{(x-x_1)(x-x_3)}{(x_2-x_1)(x_2-x_3)}$$

$$+ y_3 \frac{(x - x_1)(x - x_2)}{(x_3 - x_1)(x_3 - x_2)}$$

$$x = 5$$

$$\therefore f(5) = \frac{11(3-9)(3-12)}{(7-9)(7-12)} + \frac{17(5-7)(5-12)}{(9-7)(9-12)} + \frac{24(5-7)(5-9)}{(12-7)(12-9)}$$

$$= 30.8 + 39.6 + 12.22$$

$$= 3.92$$

$$\therefore \boxed{f(5) = 3.92}$$

$$5 > \int_{-1}^0 x e^x dx$$

Using Trapezoidal Formula.

$$\int_a^b f(x) dx = \frac{b-a}{2} \left[f(x_2) + 2 \sum f(x_1) + f(x_n) \right]$$

$$\text{where } h = \frac{b-a}{n} = \frac{0 - (-1)}{6} = \frac{1}{6}$$

$$f(x) = x e^x$$

⑩ Now

$$f(x_0) = 2.718$$

$$f(x_1) = ~~8.0~~ 3.638$$

$$f(x_2) = 4.873$$

$$f(x_3) = 6.450$$

$$\int_1^2 x e^x dx = \frac{1}{12} \times (2.718 + 2(6.637 + 4.873 + 6.450 + 8.409 + 10.735) + 14.778)$$

$$= \frac{1}{12} \times (2.718 + 68.298 + 14.778)$$

$$= \frac{1}{12} (85.794)$$

$$\approx 7.15$$

thus, the ^{evaluated} value ≈ 7.15

3)	n	:	0	1	2	3	4	5
	$f(n)$	:	0	?	8	15	?	35

the missing terms are

let's assume $f(n)$ is a quadratic polynomial.

$$f(n) = an^2 + bn + c$$

$$\text{where, } a=1, b=2, c=0$$

$$\text{for } n=0, f(n)=0$$

$$n=1, f(n)=3$$

$$n=2, f(n)=8$$

$\vdots$

$$n=4, f(n)=24$$

$\therefore$ The missing terms are 3 and 24

$$f(1)=3$$

$$f(4)=24.$$